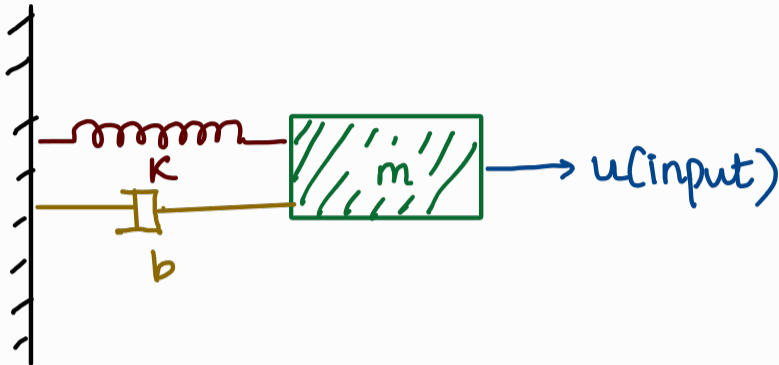


Control Systems



Mass-spring Damper

$$F_{\text{spring}} = -kx$$

$$F_{\text{damp}} = -bx$$

$$m\ddot{x} = -kx - bx + u$$

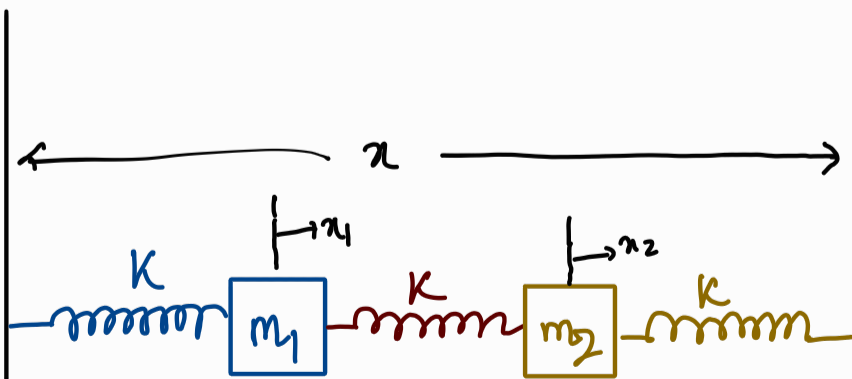
$$\Rightarrow m\ddot{x} + bx + kx = u$$

$$\frac{m\ddot{x} + bx + kx = u}{\uparrow \quad \uparrow \quad \uparrow}$$

$$\rightarrow \dot{x} = \frac{\partial x}{\partial t} \quad \left| \quad \frac{dx}{dt}\right.$$

$$\ddot{x} = \frac{\partial^2 x}{\partial t^2} \quad \left| \quad \frac{d^2 x}{dt^2}\right.$$

Question:



$$m_1 \ddot{x}_1 = -2kx_1 + kx_2$$

$$m_2 \ddot{x}_2 = -2kx_2 + kx_1$$

linear Control Theory

All terms must have a power of one

$$F(x) = cx \rightarrow F(a) = ca, F(b) = cb$$

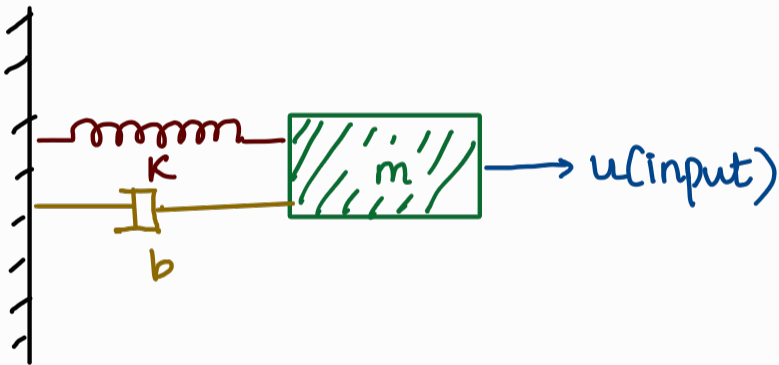
$$F(a+b) = c(a+b)$$

$$F(x) = cx + b \rightarrow F(a+b) = c(a+b) + 2b$$

linear Systems

$$\dot{X} = AX + Bu$$

Ex1:



$$X = \begin{bmatrix} \dot{x} & x \end{bmatrix}$$

$$\dot{X} = \begin{bmatrix} \ddot{x} & \dot{x} \end{bmatrix}$$

$$\dot{X} = AX + Bu$$

$$\Rightarrow \begin{bmatrix} \ddot{x} \\ \dot{x} \end{bmatrix} = A \begin{bmatrix} \dot{x} \\ x \end{bmatrix} + Bu$$

$$m\ddot{x} + b\dot{x} + kx = u \Rightarrow m\ddot{x} = -b\dot{x} - kx + u$$

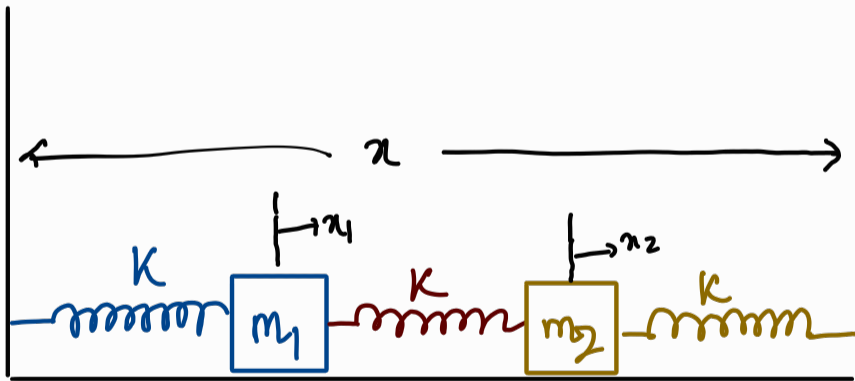
$$\Rightarrow \ddot{x} = -\frac{b}{m}\dot{x} - \frac{k}{m}x + \frac{1}{m}u$$

$$\dot{x} = 1 \times \dot{x} + 0 \times \ddot{x} + 0 \times u$$

$$\begin{bmatrix} \dot{x} \\ \ddot{x} \end{bmatrix} = \begin{bmatrix} -\frac{b}{m} & -\frac{k}{m} \\ 1 & 0 \end{bmatrix} \begin{bmatrix} \dot{x} \\ x \end{bmatrix} + \begin{bmatrix} \frac{1}{m} \\ 0 \end{bmatrix} u$$

$$A = \begin{bmatrix} -b/m & -k/m \\ 1 & 0 \end{bmatrix} \quad u = \begin{bmatrix} 1/m \\ 0 \end{bmatrix}$$

Ex 2:



$$m_1 \ddot{x}_1 = -2kx_1 + kx_2$$

$$m_2 \ddot{x}_2 = -2kx_2 + kx_1$$

$$x = \begin{bmatrix} x_1 & \dot{x}_1 & x_2 & \dot{x}_2 \end{bmatrix}$$

$$u = 0$$

$$\dot{x} = \begin{bmatrix} \dot{x}_1 & \ddot{x}_1 & \dot{x}_2 & \ddot{x}_2 \end{bmatrix}$$

$$\dot{X} = AX$$

$$\Rightarrow \begin{bmatrix} \dot{x}_1 \\ \ddot{x}_1 \\ \dot{x}_2 \\ \ddot{x}_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ -2k/m_1 & 0 & k/m_1 & 0 \\ 0 & 0 & 0 & 1 \\ k/m_2 & 0 & -2k/m_2 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ \dot{x}_1 \\ x_2 \\ \dot{x}_2 \end{bmatrix}$$

$\hookrightarrow A$

Laplace Transforms

$$F(s) = \int_0^{\infty} f(t)e^{-st} dt$$

For $f(t) = k$

$$\begin{aligned} F(s) &= \int_0^{\infty} ke^{-st} dt \\ &= -\frac{k}{s} e^{-st} \Big|_0^{\infty} \end{aligned}$$

$$= \lim_{t \rightarrow \infty} \left(-\frac{k}{s} e^{-st} \right) - \left(-\frac{k}{s} e^{-st} \right) \Big|_{t=0}$$

$$= 0 + \frac{k}{s}$$

$$= \frac{K}{S}$$

$$f(t) = K \rightarrow F(s) = \frac{K}{S}$$

$$f(t) = e^{at}$$

$$\begin{aligned} \int_0^{\infty} e^{(a-s)t} dt &= \int_0^{\infty} e^{(a-s)t} dt = \lim_{t \rightarrow \infty} \frac{1}{(a-s)} e^{(a-s)t} - \frac{1}{(a-s)} e^{(a-s)t} \Big|_{t=0} \\ &= \frac{1}{(s-a)} \end{aligned}$$

$$f(t) = t$$

$$\begin{aligned} F(s) &= \int_0^{\infty} t e^{-st} dt = \frac{t}{s} e^{-st} \Big|_0^{\infty} - \int_0^{\infty} \frac{e^{-st}}{s} dt \\ &= \frac{t e^{-st}}{s} - \frac{e^{-st}}{s^2} \\ &= \frac{1}{s^2} \end{aligned}$$

Q:- Laplace of $t^n \rightarrow$ with steps ✓

Ex:

Laplace of $\sin(\omega t)$

$$\sin \omega t = \frac{e^{i\omega t} - e^{-i\omega t}}{2i}$$

$$\mathcal{L}\left\{\frac{e^{i\omega t} - e^{-i\omega t}}{2i}\right\} = \frac{\left\{\frac{1}{s-i\omega} - \frac{1}{s+i\omega}\right\}}{2i}$$

$$= \frac{\frac{2i\omega}{s^2 + \omega^2}}{2i}$$

$$= \frac{\omega}{s^2 + \omega^2}$$

$$\cos \omega t = \frac{s}{s^2 + \omega^2}$$

$$\frac{\partial}{\partial t} = s \rightarrow \frac{\partial^2}{\partial t^2} = s^2$$

HW: $\rightarrow$ Derive these ✱

$$m\ddot{x} + b\dot{x} + kx = u$$

(Mass-Spring Damper)

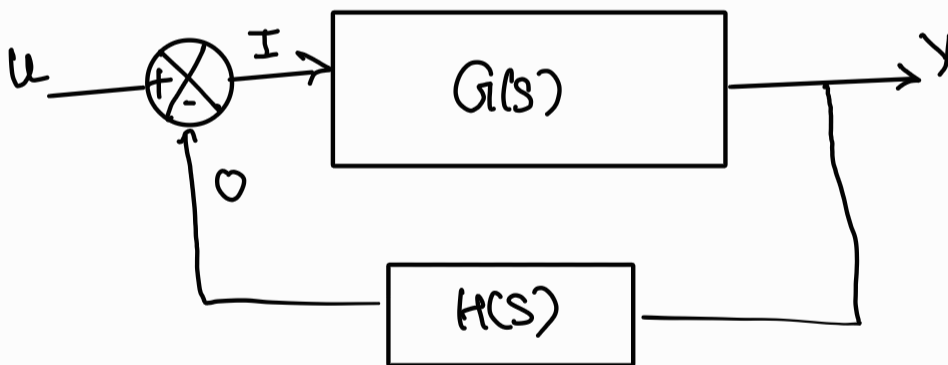
$$\Rightarrow x [ms^2 + bs + k] u$$

$$\Rightarrow a = \frac{u}{ms^2 + bs + k} = \frac{1}{ms^2 + bs + k} \cdot u$$

Open/Closed Loop Systems



Open Loop system



$$I = u - D$$

$$I \cdot G(s) = y$$

$$\Rightarrow D = I \cdot G(s) \cdot H(s)$$

$$y \cdot H(s) = D$$

$$\Rightarrow I [1 + G(s)H(s)] = u$$

$$\Rightarrow I = \frac{u}{1 + G(s)H(s)}$$

$$\Rightarrow Y = \frac{u \cdot \boxed{G(s)}}{\boxed{1 + G(s)H(s)}} \rightarrow 0$$

↓

$S \rightarrow$ Complex plane $\rightarrow$ poles (Denominator)

$S \rightarrow$ Complex plane $\rightarrow$ zeros (Numerator)

